

SONiX 32-Bit Cortex-M0 Micro-Controller

Jump To Boot ISP Example Code

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AMENDENT HISTORY

Version	Date	Description
1.0	2024/12/02	1. First version.

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1 OVERVIEW

This document provides our customers the C program examples that jump from User Code to Boot Code for ISP.

2 FEATURE

- Jump To Boot ISP example.
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.
- Connect SNLink URX/UTX to desired MCU UTX/URX.

4 EXAMPLE CODE

4.1 Initialization

Initialize systick function that MCU toggle P2.1 in systick ISR.

Initialize UART as Baud Rate 921600, data length = 8bit, odd parity and 1 stop bit.

```
55 //Toggle P2.1 in SysTick ISR
56 SysTick_Init();
57 //Initiallize UART0 setting as Baud = 921600, 8bit ,Odd parity, 1 stop bit
58 IAP_UART0_Init();
59
```

4.2 Jump to Boot Loader when receive CMD by UART

Once a specific command is received, jump to BOOT and wait for the ISP commands.

```
60 while(1)
61 {
62     if(bUART0_RecvNew == 1)
63     {
64         //If recevie IAP_GET_VER CMD means ISP start, jump to BL
65         if(bUART0_RecvFIFO[bUARTcnt++] == IAP_GET_VER)
66         {
67             //Disable all intterrupt before jump to BL
68             NVIC->ICER[0] = 0xFFFFFFFF;
69             //Jump to BL
70             Goto_Bootloader();
71         }
72         bUART0_RecvNew = 0;
73     }
74 }
75
76 }
77
78 /*****
79 * Function      : Goto_Bootloader
80 * Description   : Jump to Boot Loader address.
81 * Input        : None
82 * Output       : None
83 * Return       : None
84 * Note        : None
85 *****/
86 __asm_void Goto_Bootloader(void)
87 {
88     LDR r0, =0x1fff00c1
89     BX  r0
90 }
```

5 RESULT

The SNLink ICP Tool can execute ISP function with the sample code



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Main Office:

Address: 10F-1, NO. 36, Taiyuan Street., Chupei City, Hsinchu, Taiwan R.O.C.
Tel: 886-3-5600 888
Fax: 886-3-5600 889

Taipei Office:

Address: 15F-2, NO. 171, Song Ted Road, Taipei, Taiwan R.O.C.
Tel: 886-2-2759 1980
Fax: 886-2-2759 8180

Hong Kong Office:

Unit No.705, Level 7 Tower 1, Grand Central Plaza 138 Shatin Rural Committee Road, Shatin, New Territories, Hong Kong.
Tel: 852-2723-8086
Fax: 852-2723-9179

Technical Support by Email:

Sn8fae@sonix.com.tw